

FUTURE SCOPE

1. Expansion of Mental Health Disorder Detection

- Extend the system's capabilities to detect other mental health conditions such as anxiety, bipolar disorder, and PTSD by training on multi-disorder EEG datasets.

2. Enhanced EEG Signal Preprocessing

- Integrate advanced noise filtering and artifact removal techniques to improve signal clarity and model accuracy, especially in non-clinical environments.

3. Multimodal Data Integration

- Combine EEG data with other biomarkers (e.g., heart rate variability, facial expressions, speech patterns) for more robust and holistic assessments.

4. Remote Data Collection and Telepsychiatry Integration

- Enable integration with wearable EEG devices and telepsychiatry platforms to support remote diagnosis and consultations.

5. Personalized Depression Risk Profiling

- Develop user-specific profiles that consider individual baseline EEG patterns, lifestyle, and history to provide customized risk assessments

6. Real-Time Monitoring and Alerts

- Implement real-time EEG data processing and alert systems to notify users or caregivers of early depressive symptoms as they occur.

7. Continuous Learning and Model Adaptation

- Use reinforcement learning or continual learning algorithms to update the model based on new user data and evolving patterns in mental health.

8. Self-Help Recommendations and Mental Wellness Tips

- Incorporate evidence-based self-help strategies and mental wellness advice based on individual predictions and mood patterns.

9. Multilingual and Emotion-Sensitive Feedback System

- Provide user-friendly, emotionally aware feedback in multiple languages to support diverse populations and reduce stigma in mental health care.

10. Mobile and Cloud-Based Platform Development

- Develop a cross-platform mobile and cloud-based solution to enable easy access and storage of EEG data and prediction results, enhancing usability and reach